METRIC

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MILITARY SPECIFICATION

HEATER ASSEMBLY, COMBUSTION VEHICULAR COMPARTMENT, 8.8/17.6kW (30,000/60,000 BTU/HR), GENERAL SPECIFICATION FOR

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a forced air inlet personnel heater assembly, referred to herein as "heater" (see 6.1).

1.2 Classification. Heaters shall be of the following types and classes, as specified:

Type I	-	Heaters that must take their combustion and ventilating air from the same source ("Single Air").
Class I	-	Heaters that have a nominal rating of 8.8 kilowatts (kW) (30,000 British Thermal Units per Hour [Btu/hr]), but are also capable of operating at a nominal heating rate of 4.4 kW (15,000 Btu/hr).
Class II	-	Heaters that have a nominal maximum rating of 17.6 kW (60,000 Btu/hr), but are also capable of operating at a nominal heating rate of 8.8 kW (30,000 Btu/hr).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2540

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Type II - Heaters that can take their combustion and ventilating air from separate sources ("Dual Air").

Class II - Heaters that have a nominal maximum rating of 17.6 kW (60,000 Btu/hr), but are also capable of operating at a nominal heating rate of 8.8 kW (30,000 Btu/hr).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited).
GG-P-455	- Plates and Foils, Photographic (Photosensitive Anodized Aluminum).
TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings.
TT-T-266	- Thinner: Dope and Lacquer (Cellulose-Nitrate).
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-T-13513	- Terminal, Lug; Splice, Conductor (Electrical, Solderless for Automotive use).

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MIL-P-14105	- Paint, Heat Resisting (for Steel Surfaces).
MIL-S-22473	- Sealing, Locking, and Retaining Compounds: (Single Component).
MIL-C-22750	- Coating, Epoxy-Polyamide.
MIL-B-43436	- Band, Marker Blank.
MIL-F-45356	- Filter, Fluid, Pressure, and Filter Element, Fluid, Pressure: for Gasoline, Internal Combustion Engines, with Carburetor.
MIL-S-46163	- Sealing, Lubricating and Wicking Compounds: Thread-Locking, Anaerobic, Single-Component.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-53022	- Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.
MIL-P-53030	- Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-H-62550/1	- Heater Assembly, Combustion, Vehicular Compartment, 17.6 kW, Single Air Type (Metric).
MIL-H-62550/2	- Heater Assembly, Combustion, Vehicular Compartment, 17.6 kW, Dual Air Type (Metric).
MIL-H-62550/3	- Heater Assembly, Combustion, Vehicular Compartment, 8.8 kW, Single Air Type (Metric).
MIL-C-81562	- Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited).
MIL-M-81594	- Marking Foil, Hot Stamp Printing of Electrical Insulating Materials.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.

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MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-STD-1261	- Arc Welding Procedures for Construction Steel
MIL-STD-1275	- Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.
MIL-STD-45662	- Calibration System Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

11669705	- Box Assy, Control.
12268109	- Box Assembly, Accessory Control.
12330264	- Control Box, Electrical, Heat-Ventilator.
12345539	- Driver's Master Panel Assembly.

PURCHASE DESCRIPTIONS

ATPD 2165	- Heater Test Stand, Valcom Model VHTS-89102

(Copies of specifications, standards, drawings, and other Government documents required by the contractors in connection

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with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Performance Test Code - Application Part II of Fluid Meters Sixth (PTC) 19.5 Edition 1971 (Interim Supplement 19.5 on Instruments and Apparatus).

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 E. 47th Street, New York, NY 10017).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633	- Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel.
ASTM D439	- Automotive Gasoline.
ASTM D2156	- Smoke Density in Flue Gases from Burning Distillate Fuels, Test for.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

210-74	- Laboratory Methods of Testing Fans for Rating.

(Application for copies should be addressed to the Air Movement and Control Association, 30 W. University Drive, Arlington Heights, IL 60004).

SOCIETY OF AUTOMOTIVE ENGINEERS, INC.

SAE J726	- Air Cleaner Test Code.

(Applications for copies should be addresses to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096).

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for related associated detail specifications, specification sheets, or MS sheets), the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish heaters which shall be subjected to first article inspection (see 4.4 and 6.3). First article inspection samples, properly marked with identifying information, shall be representative of the unit to be furnished to the Government. All subsequent heaters delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein, in applicable specifications or drawings, and shall be asbestos free. Materials not specifically designated shall be suitable for use in heaters operating as specified herein without any degradation in performance of the heater (see 6.4). The heater shall be fabricated of suitable corrosion resistant materials or be treated to prevent corrosion. The heater combustion chamber, or heat exchanger, shall be made of stainless steel or other material of equivalent performance (see 4.8.1).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.4).

3.2.2 Dissimilar metals. Except where necessary to complete an electrical circuit, the compatibility of dissimilar metals shall be in accordance with MIL-STD-889 (see 4.8.1).

3.2.3 Steel components. Nonpainted steel components, other than stainless steel, shall be zinc plated in accordance with ASTM B633, type II, class FE/Zn 13 or MIL-C-81562, type II, class 5. Heat radiating surfaces may be aluminized steel (see 4.8.1).

3.2.4 Aluminum components. Aluminum components shall be coated in accordance with MIL-C-5541, class 1A (see 4.8.1).

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3.2.5 Identification plates. Nameplates shall be furnished in the configuration as specified below (see 4.8.1):

a. Material and construction shall conform to MIL-P-514, type III, composition A, class 2, or composition C. Composition C material shall conform to GG-P-455, type II, grade A, class 1. Nameplates shall be permanently attached with metal fasteners.

3.3 Design and construction. Heaters shall be constructed to the form and dimensions depicted on the applicable drawing specified by the contracting authority (see 4.8.1, 4.8.2 and 6.2). Type II heaters shall be field convertible (see 3.3.3 and 3.4.15). Other than as specified herein, the heaters shall have no provisions for adjustments of their control circuits.

3.3.1 Weight. Class I heaters shall not weigh more than 9.5 kilograms (kg) (21 pounds [lbs]). Class II heaters shall not weigh more than 17 kg (38 lbs) (see 4.8.3).

3.3.2 Electrical components and wiring. The heater electrical components and wiring shall operate and meet all environmental requirements. The wire and insulation shall be flexible to minus 54 degrees celsius (°C) (-65 degrees Fahrenheit [°F]) and be sized in accordance with the required rating. Crimped terminals shall be crimped according to MIL-T-13513. Soldering shall conform to MIL-STD-454, requirement 5 (see 4.8.1).

3.3.3 Interchangeability. All component assemblies and parts shall be functionally and dimensionally interchangeable with like component assemblies and parts previously furnished by the same manufacturer. Construction shall be such that any part, except those furnished in matched sets or for which a selective fit is specified, may be installed, replaced or adjusted without requiring modification. Heaters conforming to Type II, Class II or MIL-H-62550/2 shall be easily convertible to MIL-H-62550/1 through the use of screw drivers, hex keys or hex sockets. Once removed, no parts other than fasteners, shall be returned to the heater. The conversion of Type II heaters to Type I heaters shall not remove any protective or sensing devices needed to fulfill requirements specified herein. After conversion, dimensional conformance to MIL-H-62550/1 shall be verified (see 4.8.1).

3.3.4 Safety. In order to prevent contact burns by hot spots the contractor shall design the heater with insulation or guards. Contractors may designate a hot spot as guarded if by virtue of the heater installation, in all currently fielded vehicles, the hot spot is isolated from contact. The heater shall automatically shut off its fuel and enter its purge cycle in the event of overheating, loss of flame, or failure to ignite. The

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heater shall be so designed that, with proper installation, the heater exhaust gasses cannot enter the vehicle's personnel compartment (see 3.4.10). Radioactive material (see 6.9.3) shall not be incorporated into the heater assembly (see 4.8.1).

3.3.5 Sealant. All sealing threaded fittings in the heater's fuel circuit or threads utilizing a locking compound shall be sealed or locked, as applicable, with compound conforming to MIL-S-22473 or MIL-S-46163. Compound's grade shall be selected so that its strength shall not result in part damage upon disassembly for repair.

3.3.6 Air bleed. Heaters shall be equipped with an air bleed valve for purging air from the (vehicle) heater fuel line. The valve shall be operated by hand or with a 1/4 inch wrench.

3.3.7 Control circuits.

3.3.7.1 Start cycle. Upon initiating the start cycle with the heater control box (Drawing 11669705, 12268109, 12330264, and 12345539), the heater control circuits shall control the heater start cycle and, upon starting, automatically switch to run mode. If the heater fails to start in the allotted time (see 3.4.8), the heater shall automatically exit the start mode, diagnose the fault, and exhaust any fuel vapors (see 3.4.9).

3.3.7.2 Heat output. The heat output of heaters shall be manually setable (with control box 11669705, 12268109, 12330264, and drivers master panel assembly 12345539) or automatically reduce when excessive ventilating air temperatures occurs (see 3.4.6).

3.3.7.3 Self diagnostics. For class II heaters, the heaters, support system (fuel and air delivery), and repair part faults, shall be indicated by a flashing control box lamp. Faults shall be identified individually by the heater without the need for reference to additional documentation by maintenance personnel.

3.3.8 Accessibility. The air bleed valve (see 3.3.6) and fuel inlet shall be accessible without special tools and without disassembling the heater.

3.3.9 Welding. All heater parts that are exposed to combustion gases during normal operation shall be welded, as applicable, according to MIL-STD-1261, Class II.

3.3.10 Gaskets. Unless otherwise specified herein (see 3.3.5) no formed in place gaskets shall be used with heater parts that can be disassembled.

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3.3.11 Brush inspection. All electric motors, as applicable, shall have provisions for brush inspection or indication of brush wear without disassembly of the motors.

3.3.12 Designated spare part. Class I heaters only. If the contractor so wishes, he may designate the igniter as a spare part. If he does so, he shall include a spare igniter within the heater. If automatic switching to the spare igniter from the original igniter is not a feature of the heater design, the spare igniter shall be visible and accessible by no more than removal of the control cover.

3.3.13 Filters. Filters or screens used in the heater's fuel system shall not be less than 140 microns.

3.3.14 Wire identification. Each wire end or connector for each heater component shall be uniquely marked by the contractor according to MIL-B-43436 or MIL-M-81594. Marking the igniter leads is optional if they are otherwise readily identifiable.

3.3.15 Fuel compatibility. The heater shall be capable of operating on JP-5 per MIL-T-5624, JP-8 per MIL-T-83113, and DF-1, DF-2 and DF-A per VV-F-800.

3.3.16 Equipment compatibility. The heater shall be compatible with control box, 11669705, 12268109 and 12330264, heater test stand ATPD 2165, and driver's master panel assembly 12345539.

3.4 Performance. The heater shall meet the performance requirements specified herein when mounted in any position from horizontal to vertical with the air inlet level with, or higher than, the air outlet; and, except as otherwise specified, when operating in the high, and low heat modes. The heater shall perform as specified using fuel conforming to VV-F-800 as follows (whenever DF-A is specified herein, the contractor may substitute JP-8 according to MIL-T-83133):

Above minus 7°C (20°F)	type DF-2
Minus 7°C to minus 32°C (20 to -25°F)	type DF-1
Below minus 32°C (-25°F)	type DF-A

3.4.1 Fuel.

3.4.1.1 Fuel consumption. With the fuel pressure between 21 and 103 kilo Pascal (kPa) (3 and 15 pounds per square inch [psi]) at the fuel valve inlet, fuel consumption shall not exceed the amounts shown by table I (see 4.8.4.1).

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TABLE I. Maximum fuel flow.

Control box heat level setting	g/min 1/	(lbs/min) 2/	g/min	(lbs/min)
	Class I		Class II	
High heat	22.2	(0.049)	41.2	(0.092)
Low heat	14.1	(0.031)	23.6	(0.052)

1/ g/min = grams per minute. 2/ lbs/min = pounds per minute.

3.4.1.2 Fuel leakage. There shall be no visible leakage from the heater fuel components with the heater operating and an inlet pressure of 103 kPa (15 psi) (see 4.8.4.2).

3.4.2 Voltage. Except where otherwise specified, the heater shall start and meet performance requirements with applied voltages of 23 to 30 volts (V) direct current (dc). At ambient temperatures below minus 32°C (-25°F) the heater shall start and run with an applied voltage of 19 V dc (see 4.8.5).

3.4.2.1 Transient voltage. The heater shall not be damaged when exposed to electrical spikes and surges described by MIL-STD-1275 for both fault free and single fault operation of vehicle electrical system (see 4.8.5.1).

3.4.3 Current. At 24 V dc, the current draw from the power source shall not exceed that specified by table II (see 4.8.6).

TABLE II. Maximum current draw.

Ambient Temperature °C (°F)	Class I Start (A) 1/	Class I Run (A)	Class II Start (A)	Class II Run (A)
Above 7.2 (45)	12.0	8.0	16.0	15.0
Below 7.2 (45)	17.0	13.0	21.0	20.0

1/ A = Amperes.

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3.4.4 Heat output. The heat output of the heater shall be as shown by table III (see 4.8.7).

TABLE III. Heater heat output.

Control box heat level setting	Class I kW (Btu/hr)	Class II kW (Btu/hr)
High heat	8.8 + 0.9 - 0.9 (30,000 + 3,000 - 3,000)	17.6 + 3.6 - 1.8 (60,000 + 12,000 - 6,000)
Low heat	4.4 + 0.9 - 0.9 (15,000 + 3,000 - 3,000)	8.8 + 1.8 - 0.9 (30,000 + 6,000 - 3,000)

3.4.5 Air discharge rate. With 24 V dc applied, the ventilating air discharge rate, based on a standard air density of 1.20 kg per cubic meter (kg/m^3) (0.075 lbs per cubic foot [lbs/ft^3]) at 101.3 kPa (29.92 inches of Mercury (in Hg)), 21°C (70°F), into the specified restriction shall be as specified in table IV (see 4.8.8).

TABLE IV. Ventilating and combustion air circuits minimum restrictions.

Minimum ventilating air flow rate	Air circuits minimum restrictions Pa 3/ (in H_2O) 4/			
	Class I		Class II	
kl/min 1/ (scfm) 2/	Ventilation	Combustion	Ventilation	Combustion
3.26 (115)	0.0 (0.00)	0.0 (0.00)	N/A	N/A
2.27 (80)	187 (0.75)	187 (0.75)	N/A	N/A
(low heat)				
5.38 (190)	N/A	N/A	0.0 (0.00)	0.0 (0.00)
4.81 (170)	N/A	N/A	448 (1.80)	236 (0.95)
(high heat)				
6.37 (225)	N/A	N/A	0.0 (0.00)	0.0 (0.00)
5.81 (205)	N/A	N/A	448 (1.80)	236 (0.95)

1/ kl/min = kiloliter per minute.
2/ scfm = standard cubic feet per minute.
3/ Pa = Pascal.
4/ in H_2O = inches of water.

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3.4.6 Temperature limit. When the heater's ventilating air circuit is restricted, the heater shall limit the discharge air temperature between 150 and 177°C (300 and 350°F). When the air restriction is removed the heater shall continue burning without requiring a return to the start cycle. When the air flow is restricted to less than 1.27 kl/min (45 scfm) (class I) or 2.55 kl/min (90 scfm) (class II) the ventilating air temperature shall be 190°C (375°F) or less, the heater shall shut off its fuel flow, enter its purge cycle, and then shut itself off (see 4.8.9).

3.4.7 Smoke density. After the heater has been operating for not longer than 7 minutes, its exhaust gases shall not exceed a smoke spot number of three (five at high heat and 23 ± 10°C [73 ± 18°F]) as defined in ASTM D2156, operation at 19 V dc optional (see 4.8.10).

3.4.8 Starting time. The starting time, measured from the time the control box run/start switch is put in the "start" position until the igniter circuit is de-energized shall not exceed 2.5 minutes at temperatures above minus 7°C (positive 20°F). At lower temperatures, starting time shall not exceed 3.5 minutes. Upon starting, the control box run lamp shall light (see 4.8.11).

3.4.9 Purging cycle. The purging time, measured from the time the heater switch is turned off until the blowers automatically shut off shall not exceed 4.0 minutes. The exhaust temperature shall not be greater than 177 °C (350 °F) when the blower shuts off (see 4.8.12).

3.4.10 Toxic fumes. With the heater operational, the combustion products shall not contaminate the ventilating air. Heat exchangers shall meet the leak requirements of MIL-STD-1184 for type II, class II components, except substitute procedure 5 for procedure 1. After completion of environment and endurance testing the heater exchanger's leakage shall not exceed 14 l/min (0.5 scfm) (see 4.8.13).

3.4.11 Durability. The heater shall meet the requirements of 3.4.1 through 3.4.10, 3.4.18, 3.4.20, and 3.4.22 for 2500 hours of operation with at least 2400 start cycles and no failures (see 4.8.14).

3.4.11.1 Reduced durability test. For the control test only, the heater shall meet the requirements of 3.4.1 through 3.4.10, 3.4.18, 3.4.20, and 3.4.22 for 260 hours with at least 300 start cycles with no failures (see 4.8.14.1).

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3.4.12 Environmental conditions. The heater shall meet the requirements of 3.4.1 through 3.4.10 after being subjected to the environmental conditions specified in 3.4.12.1 through 3.4.12.6.

3.4.12.1 High temperature.

3.4.12.1.1 Storage. The heater shall be stored at an ambient temperature of at least 71°C (160°F) without sustaining damage (see 4.8.15.1.1).

3.4.12.1.2 Operating conditions. When the source of combustion air is at least 50°C (120°F), the heater shall experience no damage. The exhaust smoke spot number shall be less than three per ASTM D2156 (see 4.8.15.1.2).

3.4.12.2 Low temperature. The heater shall experience no damage when stored at minus 54°C (-65°F) or when operated at minus 50°C (-55°F) and minus 4±3°C (25±5°F) (see 4.8.15.2).

3.4.12.3 Humidity. The heater shall experience no damage after exposure to temperature and humidity cycles specified by MIL-STD-202, method 106 (see 4.8.15.3).

3.4.12.4 Salt fog. The heater shall experience no damage after exposure to salt fog for 96 hours (see 4.8.15.4).

3.4.12.5 Shock. The heater shall experience no damage after being subjected to 40 gravity units (g) shock loading for 18 milliseconds (see 4.8.15.5).

3.4.12.6 Vibration. The heater shall experience no damage or malfunction during and after sinusoidal vibration in accordance with figure 1, including up to four resonance dwells (see 4.8.15.6).

3.4.12.7 Electromagnetic interference (EMI). The heater shall meet the requirements of MIL-STD-461A, notice 4 (CE-04 and RE-02) (narrow and broad band), and operate as specified (see 3.4.4 and 3.4.8) when exposed to electrical interference specified by CS-01, RS-03 and RS-03.1) for all operating modes including the start cycle. The single switching actions that initiates and ends the heater start cycle, do not have to meet CE-04 and RE-02 EMI requirements (see 4.8.15.7).

3.4.12.8 Dust. Class II heaters shall experience no damage while operating on combustion air with at least 3 grams per cubic meter of fine dust per SAE J726 and fuel with at least 300 grams course dust per SAE J726 for at least 200 hours. Afterwards the heater shall meet the requirements of 3.4.4 (see 4.8.15.8).

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3.4.12.9 Washdown. Class II heaters shall not be damaged when subjected to mud and water applied to their combustion air outlets at a pressure of at least 275 kPa (40 psi) (see 4.8.15.9). During the start cycle, if the heater fails to start, the control box lamp shall flash and the heater shall enter its self diagnostic mode (3.3.7.3).

3.4.13 Combustion air inlet strength. Type II only. The heater's combustion air inlet shall be capable of sustaining at least 50 kg (110 lbs) load without permanent bending, cracking, interference between moving parts, or any damage that would effect heater performance or durability (see 4.8.16).

3.4.14 Starting with fault. The heater shall not be damaged when subjected to the conditions described in a, b, and c below. For conditions b and c only, the heater shall not allow more than 0.35 cubic centimeters of fuel to flow. If the heater fails to start it shall enter its purge cycle, shut off fuel flow, and then enter its self diagnostic mode in accordance with 3.3.7.3. The heater shall be operational upon verification of this requirement (see 4.8.17).

a. Heater control box start switch in the "start" position for more than 30 minutes.
b. Open igniter circuit.
c. Shorted igniter circuit.

3.4.15 Field conversion. Type II, Class II heaters (MIL-H-62550/2) shall meet the requirements of 3.4.4 through 3.4.10 with their dual air adapters removed (3.3.3) and then again with them reinstalled (see 4.8.18).

3.4.16 Inlet over pressure. Class II heaters shall start, operate and have a combustion air flow of no more than 623 liters per minute (l/min) (22 scfm) and meet the requirements of 3.4.4 when the inlet of the combustion air circuit is subjected to at least plus 2.0 kPa (8.0 in H_2O) pressure (see 4.8.19).

3.4.17 Combustion air flow. Class II heaters shall experience no damage or cause a hazardous situation when subjected to the combustion air flow conditions described below. Upon occurrence of these conditions, the heater shall shut off its fuel flow and enter the self diagnostic mode according to 3.3.7.3 (see 4.8.20). The exhaust smoke spot number throughout shall be less than three per ASTM D2156.

3.4.17.1 Reverse flow. The heater shall be subjected to a reverse pressure of at least 1.5 kpa (6.0 in H_2O).

3.4.17.2 Reduced flow. The heater shall be subjected to a reduced combustion air flow of less than 280 l/min (10 scfm).

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3.4.18 Combustion exhaust. An operating heater shall have a combustion exhaust temperature above 260°C (500°F) (see 4.8.21).

3.4.19 Ventilation. Class II heaters only. When the heater is used in the ventilation mode the ventilation circuit shall meet the air flow rate as specified in 3.4.5. The air flow rate in the combustion circuit shall not exceed 28 milliliters per minute (ml/min) (0.001 scfm) (see 4.8.22).

3.4.20 Combustion vapor exposure. For type II heaters only. Upon addition of combustion vapors (winter grade gasoline, lacquer thinner or propane) to the heater's combustion air, there shall be no luminance combustion outside the heater (see 4.8.23).

3.4.21 Over temperature power shut off. When the heater reaches its over temperature limit (3.3.4 and 3.4.6) it shall lower the voltage at pin C of the heater's connector to less than 2 Vdc and enter the purge cycle (see 3.4.9). A class II heater shall then enter its self diagnostic mode (3.3.7.3) (see 4.8.24).

3.4.22 Fuel pressure. When the fuel pressure drops below 28 kPa (4.0 psi) the control box's lamp shall flash and Class II heaters shall enter the self diagnostics mode according to 3.3.7.3 (see 4.8.25).

3.4.23 Seal leakage. When Class II heater is attached to a duct with a diameter between 201 and 209 millimeters (mm) (7.94 and 8.20 inches) and operated on high heat the leakage shall be less than 85 l/min (3 scfm) (see 4.8.26).

3.4.24 Power loss. The heater shall not be damaged, or allow more than 0.35 cc of fuel to flow when subjected to repeated shutdown without going through its purge cycle. The heater shall not enter the start cycle or operate until the control box's "start-off-run" switch is placed in the "start" position (see 4.8.27).

3.5 Protective coating. All steel parts, except stainless steel, that are in part or totally subject to temperatures exceeding 204°C (400°F), shall be coated color Forest Green or Olive Drab, per MIL-P-14105, 38.1 - 88.9 micrometers (1.5 - 3.5 mils) dry film thickness. Surfaces shall be free of oil, grease, rust and mill scale before treatment. All other painted steel parts shall be coated as follows: Clean and treat per type I or type III of TT-C-490, prime per MIL-P-53030, or MIL-P-53022; dry film thickness 20.3 micrometers (0.8 mils) minimum. The top paint color shall be camouflage green 383, chip number 34094 of FED-STD-595, conforming to MIL-C-22750, MIL-C-46168 or MIL-C-53039. Total dry film thickness of both coats shall be between 66.0 and 114.3 micrometers (2.6 and 4.5 mils).

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3.6 Identification and marking. The heaters shall be marked in accordance with MIL-STD-130 with the following information (see 4.8.1 and 4.8.2):

a. Design activity CAGE and Part or Identifying Number (PIN).
b. Manufacturer's name, CAGE, and model number.
c. Contract number.
d. Nomenclature: Heater, Vehicle Compartment, Fuel-Fired, 17.6 kW (60,000 Btu/hr) or 8.8 kW (30,000 Btu/hr), as applicable.
e. National stock number.
f. Serial number.
g. Type II (only) heaters. Combustion air inlet adapter shall be marked as follows with letters no less than 1.25 cm (0.5 in) high by black paint conforming to MIL-C-46168 or MIL-C-53039: "DO NOT REMOVE".

3.7 Workmanship. The heater shall be free of defects such as burrs, scratches, chips, sharp edges, corrosion, scale and dirt. Workmanship shall also be in accordance with requirement 9 of MIL-STD-454 (see 4.8.1 and 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2), the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept

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defective material. Unless otherwise specified in the contract (see 6.2), the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the parameter's tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
c. Control tests (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 23 ± 10°C (73 ± 18°F).
b. Barometric pressure: 96.5 ± 6.8 kPa (28.5 ± 2 in Hg).
c. Relative humidity: 50 ± 40 percent.

4.4 First article inspection. The first article test plan shall be approved by the procuring activity prior to beginning of first article testing. Unless otherwise specified (see 6.2), the Government shall randomly select three of the first ten production heaters produced under the production contract for first article inspection. First article samples shall be numbered and inspected as specified in 4.4.1 and in table V. Approval of the first article sample shall not relieve the contractor of his obligation to supply heaters that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer. Unless otherwise specified, the listing in table V does not imply any particular test sequence. Failure of the first article to meet any of the requirements specified herein will be cause for the Government to refuse to perform surveillance on additional testing until the fault(s) have been identified and corrected to the Government's satisfaction.

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TABLE V. Classification of inspections.

Title	Requirement	Inspection	First article Sample 1	2	3	Quality conformance Acceptance 100%	Control
			Hor	Vert	Hor	Horizontal	
Materials, design and construction	3.2 thru 3.2.5 3.3.2 thru 3.3.16	4.8.1	X	X	X		X
Defects (see table VII)	3.3, 3.5, 3.6 & 3.7	4.8.2	X	X	X		X
Weight	3.3.1	4.8.3	X	X	X		X
Fuel Consumption	3.4.1.1	4.8.4.1	X	X	X	X	X
Fuel leakage	3.4.1.2	4.8.4.2	X	X	X	X	X
Voltage	3.4.2	4.8.5	X	X	X	X	X
Voltage transient	3.4.2.1	4.8.5.1	X	X	X		X
Current	3.4.3	4.8.6	X	X	X	X	
Heat output	3.4.4	4.8.7	X	X	X	X	X
Air discharge rate	3.4.5	4.8.8	X	X	X	X	X
High and over temperature operation	3.4.6	4.8.9	X	X	X	X	X
Smoke density	3.4.7	4.8.10	X	X	X	X	X
Starting time	3.4.8	4.8.11	X	X	X	X	X
Purging cycle	3.4.9	4.8.12	X	X	X	X	X
Toxic fumes	3.4.10	4.8.13	X	X	X	X	X
Durability	3.4.11	4.8.14	X	X			
Reduced durability	3.4.11.1	4.8.14.1					X
High temperature							
Storage	3.4.12.1.1	4.8.15.1.1	X	X	X		
Operation	3.4.12.1.2	4.8.15.1.2	X	X	X		X
Low temperature	3.4.12.2	4.8.15.2	X	X	X		X
Humidity	3.4.12.3	4.8.15.3		X			
Salt fog	3.4.12.4	4.8.15.4			X		
Shock	3.4.12.5	4.8.15.5	X		X		X
Vibration	3.4.12.6	4.8.15.6		X	X		X
Electromagnetic interference (EMI)	3.4.12.7	4.8.15.7			X		
Dust	3.4.12.8	4.8.15.8	X		X		
Washdown	3.4.12.9	4.8.15.9	X		X		
Combustion air inlet strength test	3.4.13	4.8.16	X	X	X		X
Starting with fault	3.4.14	4.8.17	X	X	X		X
Field conversion	3.4.15	4.8.18	X				
Inlet over pressure	3.4.16	4.8.19	X				
Combustion air flow	3.4.17	4.8.20		X	X		
Combustion exhaust	3.4.18	4.8.21	X				
Ventilation	3.4.19	4.8.22	X				
Combustible vapor exposure	3.4.20	4.8.23	X	X			X
Over temperature power shut off	3.4.21	4.8.24	X				X
Fuel pressure	3.4.22	4.8.25	X	X	X		X
Seal leakage	3.4.23	4.8.26	X				X
Power loss	3.4.24	4.8.27			X		X

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4.4.1 In-process inspections. During fabrication of the first sample production heater, an in-process inspection shall be conducted by the contractor under the surveillance of the Government inspector. Inspection shall be made at the contractor's facility prior to application of primer and paint. Processing and welding procedures, quality system, and inspection records shall be reviewed and evaluated during this inspection.

4.4.2 Production sample. After completion of inspection, the first sample production heater shall remain at the manufacturing facility, as a production sample, and be the last heater shipped on the contract, unless otherwise specified herein. The heater may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the heaters during this period. The heater subject to endurance testing shall not be used to fulfill production quantities of the contract.

4.4.3 First article inspection failure. The procuring agency shall determine whether a particular first article failure shall be treated as major or minor.

a. A major failure is a non-compliance with requirements specified herein by a functioning component, heat exchanger, or when corrective action requires a design change. As a minimum, the endurance or environmental test during which the deficiency surfaced shall be repeated. A complete first article test repeat of any or all units may be necessary at the discretion of the procuring agency.

b. A minor failure, in general, involves a non-compliance caused by a non-functioning component, a minor workmanship deficiency, or when design change is unnecessary. At a minimum, a repeat of the endurance or environmental test which caused the failure shall be required.

c. The Government shall not accept delivery of any production units until all first article requirements have been satisfied.

4.5 Quality conformance inspections (QCI). QCI shall consist of the examinations of 4.5.2 and the acceptance tests of 4.5.3. Nonconformance with any of the specified requirements in sections 3 and 5 shall be cause for rejection.

4.5.1 Lot formation. An inspection lot shall consist of all the heaters of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

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4.5.1.1 Sampling for examination. Unless otherwise specified (see 6.2), sample size for quality conformance examination shall be determined in accordance with table VI. Before sampling may be initiated, the contractor shall inspect 100% of the first 100 heaters to establish that the process average (see 6.8) falls within the control limits.

TABLE VI. Sampling inspection.

Sampling lot size	Sample size for Majors	Sample size for Minors
2 to 8	*	3
9 to 25	13	3
26 to 50	13	5
51 to 90	13	6
91 to 150	13	7
151 to 280	20	10
281 to 500	29	11
501 to 1200	34	15

* Indicates entire lot must be inspected.
NOTE: The acceptance number in all cases is zero.

4.5.2 Examination. Sample units shall be examined for the defects specified in table VII.

4.5.2.1 Failure disposition. Any examination defect shall be cause for the Government to refuse acceptance of the production quantity represented by the sample size.

TABLE VII. Classification of defects.

Category	Defect	Method of examination
Major		
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
103	Faulty workmanship affecting performance (see 3.7).	Visual
Minor		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	SIE
202	Protective coating, improper (see 3.5).	Visual
203	Faulty workmanship affecting appearance (see 3.7).	Visual
204	Identification marking, improper (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

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4.5.3 Acceptance test (100%). Subsequent to the examination of 4.5.2, each heater shall be subjected to the acceptance test specified in table V.

4.5.4 Failure disposition. Any examination defect shall be cause for the Government to refuse acceptance of the production quantity represented by the sample. Any deficiency found during, or as a result of QCI shall be evidence that all items represented by the sample are similarly deficient unless contrary evidence satisfactory to the Government is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor.

4.6 Control tests. Control tests shall be conducted on one heater from each lot of 600 units consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. Control test specimens shall be representative of the production lot being offered for acceptance. The heaters shall be subjected to the control tests specified in table V.

4.6.1 Failure. Failure of any heater to pass any of the specified tests shall be cause for the Government to refuse acceptance of the production quantity represented until action taken by the contractor to correct defects and prevent recurrence has been approved by the procurement agency.

4.7 Inspection of packaging. The preservation, packaging, packing, and marking shall be inspected to determine conformance to 5.1.

4.8 Methods of inspection. Except as otherwise specified herein, performance tests shall be conducted with the heater mounted in the positions specified in table V, at the specified heat outputs, high and low heat, (see 3.4.4), and at input voltages (measured at the power input to the control box) of 23.0 ± 0.5 V dc and 30.0 ± 0.5 V dc. Acceptance testing shall be conducted at 23.0 ± 1.0 V dc only and high heat, control box optional. The fuel and ambient temperature shall be as specified in 3.4 and the fuel pressure shall be 55 ± 7 kPa (8 ± 1 psi). Prior to each start, the heater shall be cooled for at least 10 minutes. The ventilation air outlet temperature shall be measured in a duct, within 10% of its center and 105 ± 2.5cm (42 ± 1 in) from the heater. Temperature sensor shall be shielded from radiant heating by the heater. All tests shall be conducted with heater control box part number 11669705 with a nonshielded copper wire cable assembly 3 meters to 3.5 meters (10 to 12 ft) long connecting the control box with the heater. Wire gage shall be equal or greater to that used in the control box for the same circuit. If the test set up requires a longer cable, the gage of the wire and cable length shall be changed so as to provide equal

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resistance to that of the specified cable. If the heater is rated at at 8.8 kW (30,000 BTU/HR) and igniter is a designated spare part (see 3.3.12), the igniter may be replaced only once during first article testing. Otherwise the igniter shall not be replaced and no other repairs, adjustments or maintenance may be performed upon the heater during the testing specified herein. When control lamp flashing is specified the heater shall be monitored for identification of specific support system fault causing the flashing (see 3.3.7.3).

4.8.1 Materials, design, and construction. Conformance to 3.2 through 3.2.5, and 3.3.2 through 3.3.16, shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3, 3.5, 3.6 and 3.7, shall be determined by examination for the defects listed in table VII.

4.8.3 Weight. To determine conformance to 3.3.1, the heaters shall be weighed.

4.8.4 Fuel.

4.8.4.1 Fuel consumption. To determine conformance to 3.4.1.1, the fuel flow shall be measured and a fuel pressure of 103 (+0, -7) kPa (15 [+0, -1] psi) applied to the fuel valve inlet. Repeat procedure except substitute 20.7 $\pm$ .7 kPa (3.0 $\pm$ 0.1 psi) for 103 kPa.

4.8.4.2 Fuel leakage. To determine conformance to 3.4.1.2, the heater shall be operated for at least 5 minutes at a fuel pressure of 96.5 $\pm$ 6.7 kPa (14.0 $\pm$ 1 psi). Subsequently, the heater's control box cover shall be removed and the fuel circuit examined for fuel leakage.

4.8.5 Voltage. To determine conformance to 3.4.2, applied voltage shall be measured throughout performance testing (see 4.8.14).

4.8.5.1 Transient voltage. To determine conformance to 3.4.2.1, the heater shall be subjected to spike and surge voltages described by MIL-STD-1275 as being generated during fault free and single fault operation of the vehicle electrical system. The specified voltages shall be applied at least 100 times asynchronously with the switching of any electronic device that is directly subject (i.e., without active regulation of its

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voltage supply) to the spike and surge voltages of the vehicle electrical system.

4.8.6 Current. To determine conformance to 3.4.3, heater current draw shall be measured during performance testing. Running current shall be measured after the igniter or start circuit has de-energized.

4.8.7 Heat output. To determine conformance to 3.4.4, set ventilating and exhaust back pressures as specified (see 4.8.8), with temperature sensors measuring input and output air temperatures in the duct specified by 4.8.8, and measuring ventilating air flow (see 4.8.8), the heater shall be operated until the ventilating air outlet temperature has stabilized. When testing is performed at ambient temperatures above 12.8°C (55°F), and the heater's controls limit the ventilating air outlet temperature, the control may be locked out. Locking out is only allowed when the outlet temperature reaches between 135 and 177°C (275 and 350°F). Heat output shall be calculated using the following equation:

$H = V \times d \times Cp \times T$

H = Heat output in Watts (Btu/hr).
V = Ventilating air flow rate in liters per hour (l/hr) (cubic feet per hour [cu ft/hr]).
d = Density of ventilating air in kilograms per cubic meter (kg/m^3) (pounds per cubic foot [lb/cu ft]).
Cp = Average specific heat of the ventilating air (from temperature rise across heater) in kilojoules per kilogram per degree C (Kj/kg·°C) (Btu per lb per °F), including moisture content.
T = Ventilating air temperature rise across heater (outlet temperature, in degrees C(°F), minus inlet temperature, in degrees C(°F)).

4.8.8 Air discharge rate. To determine conformance to 3.4.5, establish the necessary restrictions for ventilation air and combustion gas by flowing air as specified by table VIII.

TABLE VIII. Air circuit restrictions.

Class	Ventilating Air		Combustion Gas	
	Minimum Pressure Loss kPa (in H_2O)	Flow kl/min (scfm)	Minimum Pressure Loss kPa (in H_2O)	Flow kl/min (scfm)
I	0.19 (0.75)	2.7±.14 (94±5)	0.19 (0.75)	.311±28 (11±1)
II	0.32 (1.30)	4.9±.14 (174±5)	0.24 (0.95)	.595±28 (21±1)

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Measure the air flow and pressure drops according to AMCA 210-74 or ASME PTC 19.5. During subsequent testing no changes shall be made in these restrictions. The coefficient of heat transfer of the ventilating air duct's wall shall be less than 1.14 watts per square meter per degree C (0.20 Btu/hr per square foot per degree F). Subsequently the heater shall be operated as specified herein (see 4.8) and the ventilating air flow measured as specified to determine conformance with table IV.

4.8.9 Temperature limit. To determine conformance to 3.4.6, expose the heater to a temperature between 12.8 and 32.8°C (55 and 91°F) and measure the temperature of the ventilating air in the duct and the heater fuel flow. Operate the heater into the duct used in 4.8.8, restricting the ventilating air flow as necessary until the heater limits the air's temperature to between 150 and 177°C (300 and 350°F). Continue to operate the heater for at least an additional 30 minutes. Within a period of not less than five minutes, continuously reduce the air flow to not more than 1.27 kl/min (45 scfm) (class I) or 2.54 kl/min (90 scfm) (class II). Air temperature shall remain within specified limits above. Again, continue to operate the heater for at least an additional 30 minutes. Subsequently, the air flow shall be further restricted and the heater shall shut off its fuel flow and enter its purge cycle (see 4.8.12).

4.8.10 Smoke density. To determine conformance to 3.4.7, the heater shall be operated for not more than 7 minutes. Subsequently, without changing the heater operating condition, the smoke density of the exhaust gases shall be measured according to ASTM D2156.

4.8.11 Starting time. To determine conformance to 3.4.8, put the control box start/run switch in the "start" position for 2 ± 1 seconds. Return the switch to its "run" position. Monitor the control box lamp. Repeat this test except keep the start switch in the "start" position until the lamp lights and then switch it to the "run" position. The heater shall start and operate after both procedures.

4.8.12 Purging cycle. To determine conformance to 3.4.9, the ventilating air temperature shall be stabilized at high heat and the control box starting switch shall be switched to the "off" position. Monitor and record the time for the exhaust temperature to drop to 177°C (350°F) or for the heater blower to automatically shut off; which ever occurs first.

4.8.13 Toxic fumes. To determine conformance to 3.4.10, the heat exchanger shall be removed from the heater assembly and be tested according to MIL-STD-1184, Procedure 5, with internal pressure applied as an exception. All purpose openings (see 6.4.2) for combustion air shall be sealed and the heat exchanger

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internally pressurized at no less than 0.75 kPa (3 in H2O). Leakage shall be measured according to AMCA 210-74. This test shall be performed before initial heater assembly and upon completion of all specified tests for subject heater in Table V.

4.8.14 Durability. To determine conformance to 3.4.11, the heater shall be operated as specified by paragraphs a through h:

a. Operate the heater with the specified restrictions (see 4.8.7 and 4.8.8).

b. The "ON" time (ignite and run) shall be as specified by table IX. Heater operation shall be in cycles specified by table IX and "OFF" time (purge and cool). "OFF" time shall be at least 10 minutes. The total hours of operation shall consist of only "ON" time. "OFF" time shall not be included in compilation of operating time.

c. Install a filter conforming to MIL-F-45356, type I or II, class I in the heater's fuel line. Fuel pressure shall be as specified (see 4.8) at the filter's inlet. Measure fuel pressure at heater's inlet and test according to 4.8.25. Add fine contaminate (per MIL-F-45356) to DF-2 fuel upstream of the filter at a rate not less than 0.5 grams (0.016 ounces) per hour for a total of not less than 300 grams (9.6 ounces). Monitor control box lamp for specified flashing and Class II heater for self diagnosis. After verification of 4.8.25 replace filter and continue heater testing.

d. During each cycle (start and run portion) the applied voltage shall be dropped to less than 19 V dc for 2.5 $\pm$ 0.5 seconds.

e. During endurance the heater shall be tested as specified by 4.8.11, 4.8.12, and 4.8.25 (see 4.8.14c). When testing in accordance with 4.8.25 only one step is required.

f. After each phase of endurance testing, and still at the phase's specified temperature, the heater shall be tested as specified in 4.8.4.1, 4.8.4.2, and 4.8.5 through 4.8.10, and 4.8.21.

g. Fuel shall be at the same temperature as the ambient (test) temperature.

h. Testing shall consist of the following phases:

PHASE I - Using DF-2 fuel at ambient temperature of 23 $\pm$ 10°C (73 $\pm$ 18°F)

a. Set heater control box at low heat and operate the heater for 25 cycles with 23.0 $\pm$ 0.5 V dc applied.

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b. Repeat step A using an input voltage of 28.0 ± 0.5 V dc.
c. Reset temperature control to high heat and repeat steps "a" and "b".
d. Repeat steps "a" through "c" until 600 cycles specified by table IX are completed.

PHASE II - Using DF-1 fuel and at minus -4.0 ± 3°C (25 ± 5°F) repeat steps "a" through "d", except 300 cycles specified by table IX shall be completed and in step "b" substitute 30.0 ± 0.5 V dc in lieu of 28.0 ± 0.5 Vdc.

PHASE III - Using DF-A fuel and at minus 13 ± 3°C (55 ± 5°F) repeat steps "a" through "d", except 300 cycles specified by table IX shall be completed, and in step "a" and "b" substitute less than 19 V dc and 30 ± 0.5 V dc, respectively, for the specified voltages. At the 19 V dc condition of this phase there is no requirement to drop the voltage as specified by the basic paragraph.

PHASE IV - Repeat Phases I through III for an additional 1200 cycles.

TABLE IX. Test times for class I and class II heaters.

Heater Class	On Time 1/ (Hours)	Cycle ON Time (Minutes)	Phase
I	600	30	I
I	300	30	II
I	300	30	III
II	1900	95	I
II	300	30	II
II	300	30	III

1/ Total time accumulated (including repeat of Phases specified by Phase IV) at specified test conditions.

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4.8.14.1 Reduced durability. For control test only, the durability testing (see 4.8.14) shall be reduced 260 hours and 300 start cycles. Operation shall consist of on-time as specified by table IX for the applicable class. Off-time shall be 10 minutes. The test shall consist of 100 cycles each of phases I, II and III. Addition of dust not required.

4.8.15 Environmental. Before testing the heater shall be operated for not less than 30 minutes. Environmental testing as applicable shall be performed before endurance testing. After the completion of environmental testing as specified in 4.8.15.1 through 4.8.15.9, the heater shall be subjected to the tests specified in 4.8.4 through 4.8.13.

4.8.15.1 High temperature.

4.8.15.1.1 Storage. To determine conformance to 3.4.12.1.1, the heater shall be tested as specified by MIL-STD-202, method 108, condition B at no less than 71°C (160°F). The heater shall be positioned as specified in table V. After storage, the heater temperature shall be allowed to reach a temperature of 23 ± 10°C (73 ± 18°F), and the heater shall be tested as specified in 4.8.4 through 4.8.12 using DF-2 fuel.

4.8.15.1.2 Operating conditions. To determine conformance to 3.4.12.1.2 the heater shall be operated for a minimum of one hour at high heat using combustion air at a minimum of 49°C (120°F). The operating exhaust gas smoke density shall be measured according to ASTM D2156.

4.8.15.2 Low temperature. To determine conformance to 3.4.12.2, the heater shall be tested in accordance with MIL-STD-810, method 502, procedure I. The heater shall be preconditioned by operating with DF-A fuel. The heater, positioned as specified in table V, shall be stored at not greater than minus 54°C (-65°F) for at least 12 hours. After storage, the heater temperature shall be allowed to reach a temperature of minus 48 ± 3°C (55 ± 5°F) and the heater shall be tested as specified in 4.8.4 through 4.8.12 using DF-A fuel. Repeat the test using DF-2 fuel at minus 4 ± 3°C (25 ± 5°F).

4.8.15.3 Humidity. To determine conformance to 3.4.12.3, the heater shall be tested as specified in MIL-STD-202, method 106.

4.8.15.4 Salt fog. To determine conformance to 3.4.12.4, the heater shall be tested in accordance with MIL-STD-202, method 101, condition A.

4.8.15.5 Shock. To determine conformance to 3.4.12.5, the heater, with no fuel applied, shall be tested as specified by

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MIL-STD-202, method 213, condition A, except the pulse shall have an amplitude of 40 g's and a duration of 18 milliseconds. It shall be mounted at the strap space locations shown by the applicable drawing.

4.8.15.6 Vibration. To determine conformance to 3.4.12.6, the heater shall be rigidly mounted at the strap space locations specified by the applicable drawing and subjected to vibration as specified by MIL-STD-202, method 204, condition A, using the curve shown by figure 1, in lieu of specified vibration level. During testing, heater shall be started, purged, operated and monitored for intermittent operation. A resonant frequency search shall be conducted along each axis over a range of 5 to 500 Hertz (Hz). The test schedule shall include 13-1/3 minutes dwell time at each resonance. If more than four resonances are present, the most severe shall be used. The total cycling time per axis shall be 80 minutes including the resonance dwell time. Total cycle time shall be equally divided between operation at low and high heat.

4.8.15.7 Electromagnetic interference. To determine conformance to 3.4.12.7, the heater shall be subject to the tests specified by MIL-STD-462, notice 3. While it is exposed to the specified interference, it shall be operated as specified by 4.8.7 and 4.8.11. Starting and operating the heater may be simulated by eliminating fuel combustion. If this simulation is used, the contractor's test plan shall describe completely how he intends to simulate the heater's operation and demonstrate that his simulation is adequate to ensure representative heater operation during the specified testing.

4.8.15.8 Dust. To determine conformance to 3.4.12.8, a class II heater shall be operated at 28.0 $\pm$ 0.5 vdc at both heats for at least 100 hours at each heat. During operation fine dust shall be added to the heater's combustion air at the specified rate in accordance with MIL-STD-810, method 510. Repeat the above test except add course dust to the heater's fuel at a rate of not less than 1.5 grams (0.048 ounces) per hour, for a total of not less than 300 grams (9.6 ounces). Test the heater according to 4.8.7 upon completion of the dust testing.

4.8.15.9 Washdown. To determine conformance to 3.4.12.9, a class II heater shall be subjected to the following steps:

Step 1: Test set up. Horizontally mount the heater. Horizontally orient the combustion air outlet. Attach a stand pipe to the heater's combustion air outlet with adapter(s) as required. The stand pipe and the adapters shall be shaped to direct the test material (a suspension of test grit in water) to the heater's combustion air outlet. The stand pipe and adapter shall be at least 30 cm (12 inches) high with a volume of at

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least 4 liters (one gallon). The combustion air inlet may be sealed to prevent loss of test material.

Step 2: Test material. The test grit shall consist of 70 percent AC course (per SAE J726) and 30 percent 80 grit aluminum oxide. The test material shall be in suspension of at least 125 grams (4 ounces) test grit and 4 liters (one gallon) of water. The test grit shall be maintained in suspension as it is added to the set up's stand pipe.

Step 3: Procedure. Fill the stand pipe with test material to within one centimeter (0.5 inches) of the pipe's top (add water, as necessary, stirring to maintain the suspension). Simultaneously apply at least 275 kPa (40 psi) to all of the mixture's top surface and remove the seal from the combustion air inlet. Maintain pressure for at least three to five seconds. Wait for 30 minutes and attempt to start the heater. Monitor the control lamp for the specified flashing (see 3.3.7.3). Repeat the above starting attempt procedure at least three more times. Remove the heater from the test set up and drain it for less than 30 seconds. Subsequently, operate the heater at high and low heats for at least two hours at each heat at 28.0 $\pm$ 0.5 V dc.

Step 4: Reseal the combustion air inlet if necessary (Step 1) and repeat step 3 for at least four more times."

4.8.16 Combustion air inlet strength. Type II only. To determine conformance to 3.4.13, the heater shall be suspended with a suitable strap (less than 2.5 cm [1.0 inch] wide) from each of the inlet tubes listed below. Weights shall be attached so that the total suspended weight (heater plus weights) shall be at least 50 kilograms (110 lbs). After testing the heater shall not be permanently deformed and it shall be capable of normal operations. There shall be no evidence of interfering parts, such as vibration, slow motor speeds, or sounds of rubbing parts. The heater shall be tested at the following combustion air inlet diameters:

a. 38.10 mm (1.50 in) diameter. Measure the torque necessary to rotate the adapter through its specified range. Support heater as specified. Remove the heater from the test setup. Inspect the heater and rotate the combustion air inlet. The inlet shall rotate freely with no binding and the torque to rotate it shall not change by more than 20%.

b. 43.18 mm (1.7 in) max diameter. Support heater from center (within 15 percent) of tube as specified..

4.8.17 Starting with fault. To determine conformance to 3.4.14, the heater shall be subjected to the tests a, b, and c

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below. These test shall be completed prior to endurance and shock testing.

a. Move the control box's start-run switch to the start position and hold it there for at least 30 minutes. The heater shall start and operate upon completion of this test.

b. Remove the igniter from the heater and leave the heater's igniter circuit open. Move the control box's start-run switch to the start position and hold for 30 minutes. Measure and record the total fuel flow delivered to the heater.

c. Repeat the test from item b with the igniter circuit shorted out using a copper wire of at least 10 gage and not longer than 30 cm (10 inches). Measure and record the total fuel flow to the heater.

d. Re-install the igniter and the heater shall be operable.

4.8.18 Field Conversion. To determine conformance to 3.4.15, Type II, Class II heaters (MIL-H-62550A/2) heaters shall have their dual air adapters removed with only the specified tools (3.3.3) and be subjected to the tests specified in 4.8.7 through 4.8.13. The adapters shall be reinstalled (use only removal tools) and the heaters shall again be subjected to these same tests.

4.8.19 Inlet over pressure. Class II heaters only. To determine conformance to 3.4.16, 2.0 kPa (8 in H_2O) pressure shall be applied to the heater's combustion air inlet. The combustion air flow shall be measured according to AMCA 210-74 or ASME PTC 9.5. Subsequently the heater shall be started, operated and tested as specified in 4.8.7.

4.8.20 Combustion air flow. Class II heaters only. To determine conformance to 3.4.17, the heater shall be subjected to the following tests prior to beginning the endurance test.

4.8.20.1 Reverse flow.

a. Apply at least 1.5 kPa (6.0 inches H2O) to the heater's exhaust. Attempt to start the heater. Monitor the control box lamp. Repeat this start test three times.

b. Operate the heater and apply 1.5 kPa (6.0 inches H2O) to the heaters exhaust for 45 to 60 seconds. Remove the pressure from the exhaust for 3 to 4 minutes. Reapply the pressure to the exhaust for at least another 5 minutes. Monitor the fuel flow, the control box lamp, and the combustion air inlet temperature for the above procedure. The combustion air inlet temperature shall be measured with in 3 cm (1.2 inches) of the inlet. Record the heaters reaction to the above parameters, whether the heater

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enters the purge cycle and the time it takes to enter the purge cycle. Note also the duration of the control box lamp flashing. Repeat this operating test three times.".

4.8.20.2 Reduced flow.

a. Reduce air flow to less than 280 l/min (10 scfm) measured per ASTM 210-74 or ASTM PTC 9.5. Attempt to start the heater. Note if the heater starts and the control box lamp condition. Repeat this start test 10 times.

b. Operate the heater for at least 1 hour. Slowly reduce the air flow to less than 280 l/min (10 scfm) measured per ASTM 210-74 or ASTM 9.5. During flow reduction, measure the smoke dot (per ASTM D2156) and the control box lamp. Record these conditions and the time and duration when the heater enters the purge cycle. Repeat this operating test 10 times.".

4.8.21 Combustion exhaust. To determine conformance to 3.4.18, the heater combustion exhaust temperature shall be measured with the heater operating. The measurements shall be made, as a minimum, during the endurance and performance tests.".

4.8.22 Ventilation. Class II heaters only. To determine conformance to 3.4.19, power shall be applied to pin E of its connector and air flow in the combustion and ventilation circuits measured according to AMCA 210-74 or ASME PTC 19.5.

4.8.23 Combustible vapor exposure test. For Type II heaters only. To determine conformance to 3.4.20, the heater's adapters shall be removed and the heater operated with a combustible mixture consisting of a saturated vapor mixed with air delivered to the combustion air inlet. The saturated vapor shall be generated by a chamber as specified herein. After the removal of the adapters and before the exposure to the combustible mixture, the air flow shall be measured. During the exposure to the combustible mixture, its flow shall be measured again. The air flow shall not vary by more than 5% from the previously measured air flow. The heater shall be subjected to the following steps:

a. Heater configuration. Remove the heater adapters to convert to a Type I configuration. Deliver a combustible mixture to the combustion air inlet and allow for observation of the luminous combustion from outside its body.

b. Chamber design and operation. The chamber shall be closed but for means of adding and removing the air, saturated vapor and draining spent test fluid. It shall be designed to keep the air from bubbling through the test fluid and to prevent fluid carry over. The test fluid shall wet the chamber's walls and shall not be recirculated or resprayed through the chamber. Sufficient

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fluid shall be sprayed into the chamber to form a steady stream of spent test fluid in the drain. While testing, saturation of the vapor shall be shown by warming the chamber's incoming air, cooling the chamber's exiting air within one degree °C (two degree °F) of the combustion air to condense a portion of the vapor and continuously remove the condensate.

c. Saturated vapor. The saturated vapor shall be generated by the chamber from lacquer thinner conforming to TT-T-266 and repeated with winter grade gasoline conforming to ASTM D439.

d. Combustion air. The heater's combustion air shall be mixed with a combustible gas (saturated vapor or propane, as applicable) to give a test mixture. An increasing amount of combustible gas shall be added to the combustion air linearly over a period of at least 20 minutes, starting at less than one percent combustible gas and ending with 100 percent saturated vapor or 25 percent propane.

4.8.24 Over temperature power shut off. To determine conformance to 3.4.21, the voltage at pin C of the heater's connector shall be measured. The heater shall then be operated and the ventilation air flow restricted until the heater shuts off the fuel flow and enters the purge cycle (see 4.8.12). After the heater enters the purge cycle the voltage at pin C shall be measured.

4.8.25 Fuel pressure. To determine conformance to 3.4.22, perform the following two steps:

Step 1. Before entering the start cycle, lower the fuel pressure to 27.6 ± 0.7 kPa (4.0 ± 0.1 psi). Start the heater and monitor the control box lamp. Shut the heater off. Raise the fuel pressure to the specified value (55 ± 7 kPa [8 ± 1 psi]). Start the heater and monitor the control box lamp. The lamp shall not flash and Class II heaters shall not indicate a fault.

Step 2. Repeat Step 1 except the heater shall have operated for at least an hour prior to exposure to low fuel pressure.

4.8.26 Seal leakage. Class II heaters only. To determine conformance to 3.4.23, the heater's ventilation air outlet shall be installed in a duct of a maximum diameter of 201 mm (7.94 inches) and operated on high heat at the specified restriction (see 4.4.8). Air leaking past the clearance between the heater and the duct shall be collected by a plenum sealed between them. The plenum's pressure shall be measured and held within 0.012 kPa (0.05 in H_2O) of the heater's ventilation air inlet. The air flow leaving the plenum shall be measured according to AMCA 210-74 or ASME PTC 19.5. The test shall be repeated except the duct shall have a minimum diameter of 209mm (8.20 inches).

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4.8.27 Power loss test. To determine conformance to 3.4.24, operate the heater for at least 95 minutes while monitoring the fuel flow. Remove the power from the heater control box. Measure the time required for the fuel to stop flowing and the amount of fuel that flowed after power loss. Wait at least 1 hour. Restore power to the control box (the heater should not start). Move the start-off-run switch to the start position (the heater should start). Repeat this power loss test 10 times.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Application information. The heater furnished under this specification is intended to supply warm air to the crew compartment of military vehicles. In order to obtain heater durability in these vehicles that heaters meeting this specification have demonstrated, the following installation guidelines should be followed. Otherwise, their durability may be unsatisfactory.

6.1.1 Restriction. The restriction of the heater air circuits should not exceed those specified by this specification (see 3.4.5).

6.1.2 Fuel. Heater fuel should not contain solid contaminates larger than 30 micron. Filters conforming to MIL-F-45356, Type I or II, Class I provide adequate protection and capacity. Filters that require frequent cleaning are not effective. The best source of heater fuel is downstream of the engine fuel filter(s).

6.1.3 Ventilator. The heater should not be used solely as a ventilating device or fan. The heater's (fan motor) life is consumed performing a task that can be much more cheaply performed by a dedicated fan.

6.1.4 Fuel pressure. Fuel pressure should be between 21 and 103 kPa (3 and 15 psi).

6.1.5 Combustion air. Combustion air should be free of dust, sand, and/or water. Otherwise the burner will become clogged with debris and the heater will fail to start. In most

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vehicles, the air taken from the crew compartment is adequate for combustion purposes. During military operations, crew compartments are free of liquid water, and dust concentrations of 0.009 g/cubic ft of AC fine dust equivalent have been measured there.

6.1.6 Fuel line. The internal diameter of the fuel lines for these heaters should be at least one half inch. This size, while not needed for fuel temperatures above minimum specified (see 3.4), helps fuel flow when the temperature is below the cloud point of the fuel in the heater's fuel line.

6.1.7 Heater power. Electric power for the heater should be taken from the "hot" side of the vehicle's master power switch. Severe damage happens to heaters when they lose power before the completion of their purge cycle.

6.1.8 Differential restriction. The vehicle's external inlet and outlet for the heater's combustion air circuit should be located so that between the openings there is no actual or potential for differential pressures greater than 2 in water.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of applicable drawing (see 3.3). Heaters conforming to Type I, Class II or MIL-H-62550/1 should not be ordered. In their lieu, heaters conforming to Type II, Class II or MIL-H-62550/2 should be ordered. The latter heaters are field convertible (see 3.3.3 and 3.4.15) to the former.
c. If first article is required (see 3.1).
d. If responsibility for inspection and the place of inspection is other than the contractors (see 4.1).
e. Arrangements for first article inspection, approval of test results, and disposition of first article (see 4.4).
f. Government's rights and conditions for waiving first article inspection (see 6.3).
g. Selection of applicable level and packaging requirements (see 5.1).
h. Sampling table, if other than as specified (see 4.5.1.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the first article samples should be first article sample, preproduction sample, initial production sample, first production item, or standard production item from the contractor's current inventory. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of

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first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.1).

6.4.1 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or by product material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4.2 Purpose opening. For the purpose of this specification, a purpose opening is an opening used to access an otherwise blocked feature.

6.5 Supersession data.

Superseded Part	New Part (MIL-H-62550)
M46792/2	Type I, Class I
7748716	Type I, Class II
11601809	Type I, Class II
11669489	Type II, Class II
11669489-1	Type II, Class II
11669490	Type II, Class II
11669490-1	Type II, Class II
11669496	Type I, Class II

6.6 Critically. This is a critical item.

6.7 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Military Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable

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Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

6.8 Process average. Sampling may be initiated if the process average value for the first 100 heaters inspected is less than 1% for major defects and less than 4% for minor defects. The process average should be calculated as follows:

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of heaters inspected}} \times 100\%$$

6.9 Definition.

6.9.1 Fault. For the purpose of this specification a fault is any condition that would prevent the heater from starting, delivering required heat, or operating under conditions that would shorten its life.

6.9.2 Heat exchanger. The heat exchanger, for the purpose of this specification, is the part of the heater that is exposed to combustion products during normal operation. System faults involving reversal of air flow (see 3.4.17.1) is not normal operation.

6.9.3 Radioactive material. Radioactive material for the purpose of this specification, is any material with a specific activity greater than 0.002 microcurie per gram or with an activity greater than 0.01 microcurie for any component.

6.10 Subject term (key word) listing.

diagnostics, self
air flow
ventilation
over pressure
smoke density
combustion vapor

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 2540-A034)

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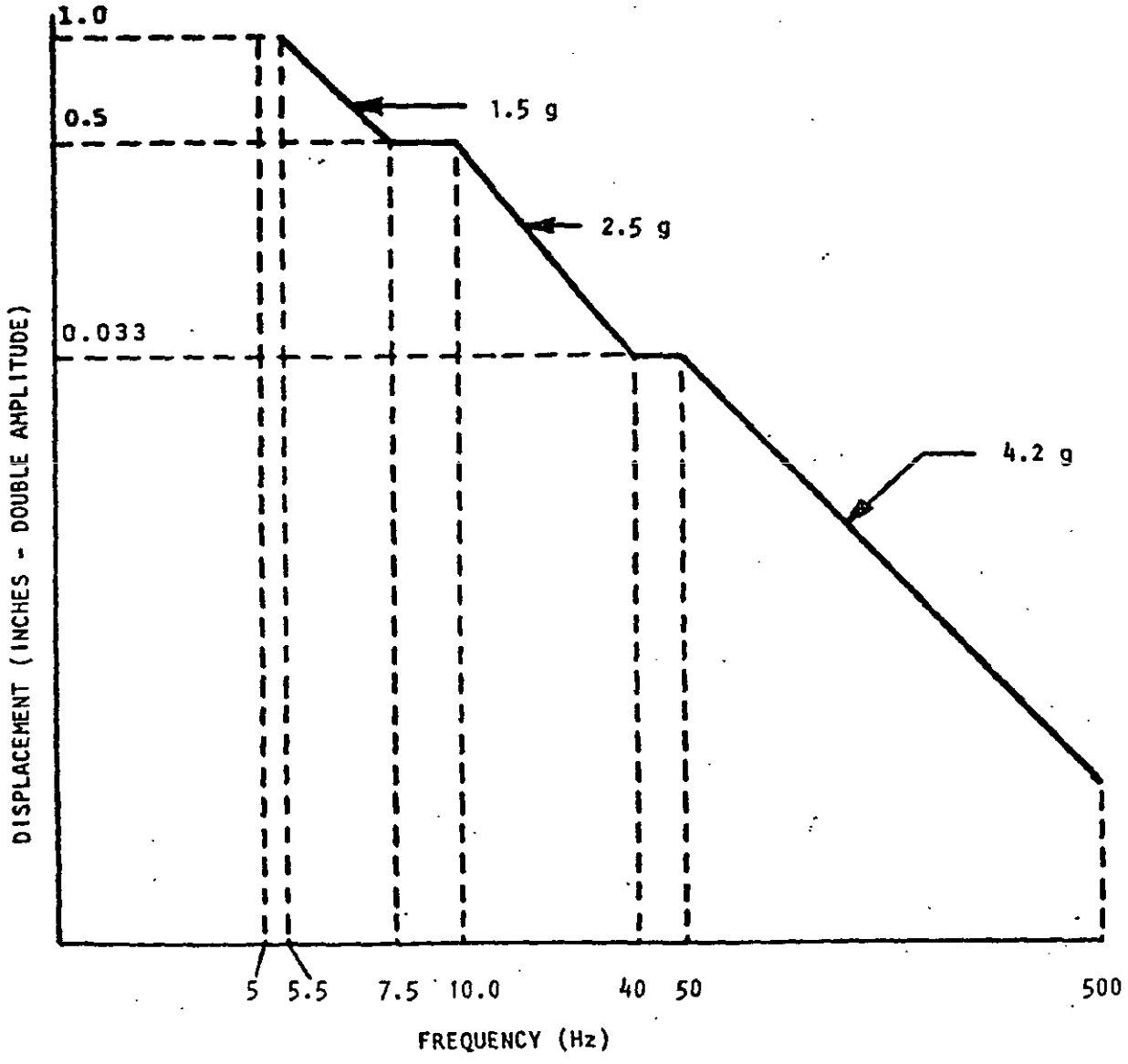


FIGURE 1. Vibration curve.

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